

SIKANUA

Nutrition & Fitness Recommendation System

Internal Working Model Document

Confidential — Engineering & Product Teams

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1. Introduction

SIKANUA is an intelligent, adaptive nutrition and fitness recommendation system built on a layered AI architecture. Its core mission is to translate a user's biometric and health profile into a personalised, evidence-based plan that covers both diet and physical activity — and to continuously refine that plan through structured weekly review cycles.

This document defines the full internal working model: the data inputs the system accepts, the AI reasoning layers it applies, the logic for plan generation, the 7-day review protocol, and the guardrails that keep recommendations safe and equitable.

Design Principle: SIKANUA treats every user as a unique clinical case. Static one-size-fits-all plans are explicitly excluded. The system is designed to be as useful for a sedentary office worker managing pre-diabetes as it is for an active athlete seeking peak performance.

1.1 Document Scope

- Target audience: product engineers, ML engineers, clinical reviewers, and QA teams.
- This document does not replace clinical guidelines; it describes how SIKANUA interprets and applies them.
- References to 'the model' mean the ensemble of AI components described in Section 4.

1.2 System Vision

SIKANUA sits at the intersection of four AI types drawn from the foundational taxonomy:

AI Type	Role in SIKANUA	Example Output
Diagnostic / Descriptive	Understand the user's current state from intake data	BMI classification, macro gap analysis
Predictive	Forecast outcomes under different plan choices	Projected weight change, energy availability
Prescriptive	Recommend the optimal plan and next actions	7-day meal plan, workout schedule, supplement flags
Limited Memory	Learn from weekly reviews to refine future plans	Adjusted macros after week-1 adherence report

2. User Intake Data Model

Before any recommendation is generated, SIKANUA collects a structured intake profile. Every field is classified by tier — Tier 1 fields are mandatory at signup; Tier 2 fields are prompted progressively; Tier 3 fields are optional and typically provided by connected health devices or a clinician.

2.1 Biometric Inputs

Field	Type	Tier	Notes
Age	Integer (years)	1	Affects BMR formula selection
Biological Sex	Enum: M / F / Intersex	1	Used for hormonal baseline; separate from gender identity
Gender Identity	Free text / Enum	2	Stored for inclusive communication only; not used in calculations
Height	Float (cm or ft/in)	1	Unit preference stored per user
Weight	Float (kg or lbs)	1	Converted to kg internally
BMI (computed)	Float	—	BMI = weight(kg) / height(m) ² ; displayed, not entered
Body Fat %	Float	2	Optional; improves lean mass estimates
Waist Circumference	Float (cm)	2	Cardiometabolic risk proxy
Hip Circumference	Float (cm)	2	Waist-to-hip ratio calculation
Resting Heart Rate	Integer (bpm)	3	From wearable or manual entry
Blood Pressure	Systolic / Diastolic (mmHg)	3	Flags hypertension tier for plan constraints
Blood Glucose (fasting)	Float (mmol/L or mg/dL)	3	Diabetes / pre-diabetes flag

2.2 Health Condition Flags

Health conditions are encoded as a boolean flag set plus free-text notes. Each active flag triggers a constraint layer in the prescriptive engine (see Section 4.3). Flags are grouped into five clinical domains:

2.2.1 Metabolic Domain

- Type 1 Diabetes
- Type 2 Diabetes
- Pre-diabetes / Insulin Resistance
- Hypothyroidism / Hyperthyroidism
- Polycystic Ovary Syndrome (PCOS)
- Metabolic Syndrome

2.2.2 Cardiovascular Domain

- Hypertension (Stage 1 or 2)
- Coronary Artery Disease
- Heart Failure (compensated)
- Hyperlipidaemia / Dyslipidaemia
- History of Stroke or TIA

2.2.3 Musculoskeletal Domain

- Osteoporosis / Osteopenia
- Rheumatoid or Osteoarthritis
- Chronic Lower Back Pain
- Post-surgical rehabilitation flag
- Injury flag (active / recovering)

2.2.4 Gastrointestinal & Renal Domain

- Irritable Bowel Syndrome (IBS)
- Inflammatory Bowel Disease (Crohn's / Colitis)
- Celiac Disease
- Chronic Kidney Disease (CKD, Stage 1–4)
- Lactose Intolerance
- GERD / Acid Reflux

2.2.5 Mental Health & Neurological Domain

- Diagnosed eating disorder (recovery flag — restricts caloric floor parameters)
- Anxiety / Depression (affects exercise intensity guidance)
- Epilepsy (flags exercise safety constraints)

Safety Rule: If any Tier-3 clinical condition is flagged (CKD Stage 3+, active eating disorder, cardiac history), the system mandates a clinician review before activating a plan. The user receives a holding message and is placed on a supervised queue.

2.3 Nutritional Requirements & Preferences

Input	Description
Dietary Style	Omnivore, Vegetarian, Vegan, Pescatarian, Keto, Mediterranean, Halal, Kosher, etc.
Food Allergies	Multi-select: peanut, tree nut, shellfish, fish, soy, wheat, egg, milk
Food Dislikes	Free-text list; used to filter meal alternatives
Preferred Cuisines	Multi-select: West African, Asian, Mediterranean, Latin American, etc.
Cooking Skill Level	Enum: None / Beginner / Intermediate / Advanced
Meal Prep Time Available	Enum: <15 min / 15–30 min / 30–60 min / 60+ min
Number of Meals Per Day	Integer: 2 – 6

Input	Description
Budget Category	Enum: Low / Moderate / High — affects ingredient choices
Alcohol Consumption	Enum: None / Occasional / Regular
Water Intake (current)	Float (litres/day) — used as hydration baseline
Supplement Use	Free-text; checked against plan to avoid duplication

2.4 Physical Activity Profile

Input	Description
Current Activity Level	Sedentary / Lightly Active / Moderately Active / Very Active / Athlete
Exercise History	None / <6 months / 6–24 months / >2 years
Preferred Exercise Types	Multi-select: Strength, Cardio, HIIT, Yoga, Swimming, Sports, etc.
Equipment Access	Enum: None / Home Equipment / Full Gym / Outdoor Only
Available Training Days/Week	Integer: 1 – 7
Session Duration Available	Enum: <20 min / 20–45 min / 45–75 min / 75+ min
Mobility / Flexibility Issues	Boolean + free-text notes
Primary Fitness Goal	Weight Loss / Muscle Gain / Endurance / Flexibility / General Health / Athletic Performance

2.5 Goal Parameters

- Primary Goal: selected from the fitness goal list above.
- Target Weight (optional): user-specified; system validates against safe minimum BMI thresholds.
- Target Timeframe: expressed in weeks; used to compute required weekly deficit or surplus.
- Priority Balance: slider input — Diet-focused vs. Exercise-focused (affects how caloric targets are distributed).
- Motivation Style: Strict / Flexible / Social / Gamified — affects how reminders and nudges are framed.

3. Baseline Calculation Engine

Before the AI recommendation layers engage, a deterministic calculation engine computes the user's energy and macronutrient baselines. These outputs are the numerical ground truth that all subsequent AI layers operate on.

3.1 BMI Classification

BMI Range	Classification	System Action
< 16.0	Severe Underweight	Block plan; escalate to clinical queue
16.0 – 18.4	Underweight	Warn; require clinician clearance for weight-loss goals
18.5 – 24.9	Normal Weight	Standard plan generation permitted
25.0 – 29.9	Overweight	Standard plan; weight-loss path available
30.0 – 34.9	Obese Class I	Standard plan; moderate caloric restriction
35.0 – 39.9	Obese Class II	Medical review recommended (non-blocking flag)
≥ 40.0	Obese Class III	Clinical queue; plan restricted until cleared

3.2 Basal Metabolic Rate (BMR)

SIKANUA uses the Mifflin-St Jeor equation as the primary BMR calculator due to its validated accuracy across diverse adult populations:

For Males: $BMR = (10 \times \text{weight_kg}) + (6.25 \times \text{height_cm}) - (5 \times \text{age}) + 5$
For Females: $BMR = (10 \times \text{weight_kg}) + (6.25 \times \text{height_cm}) - (5 \times \text{age}) - 161$

If body fat percentage is available (Tier 2), the Katch-McArdle formula is used instead for improved accuracy:

$BMR \text{ (Katch-McArdle)} = 370 + (21.6 \times \text{Lean Body Mass in kg})$

3.3 Total Daily Energy Expenditure (TDEE)

TDEE is derived by multiplying BMR by the user's Physical Activity Level (PAL) multiplier:

Activity Level	PAL Multiplier	Description
Sedentary	× 1.2	Little or no exercise; desk job
Lightly Active	× 1.375	Light exercise 1–3 days/week
Moderately Active	× 1.55	Moderate exercise 3–5 days/week
Very Active	× 1.725	Hard exercise 6–7 days/week
Athlete / Extra Active	× 1.9	Very hard exercise, physical job

3.4 Caloric Target Computation

The caloric target is the TDEE adjusted by the user's primary goal:

Goal	Caloric Adjustment
Weight Loss (Standard)	TDEE – 500 kcal/day (approx. 0.45 kg/week loss)
Weight Loss (Aggressive)	TDEE – 750 kcal/day (max; never below 1200 F / 1500 M)
Maintenance	TDEE (no adjustment)
Muscle Gain (Lean Bulk)	TDEE + 250 kcal/day (minimised fat gain)
Muscle Gain (Standard Bulk)	TDEE + 500 kcal/day
Athletic Performance	TDEE + periodisation adjustment (training phase dependent)

Hard Floor Rule: No plan may set a daily caloric target below 1,200 kcal for female users or 1,500 kcal for male users. If the computed target falls below this floor, the system locks the target at the floor value and adjusts the projected timeframe accordingly, notifying the user.

3.5 Macronutrient Distribution

Macronutrient targets are computed as percentages of total caloric target, then adjusted by health condition flags:

Goal / Profile	Protein	Carbohydrate	Fat
General Health / Maintenance	20–25%	45–55%	25–30%
Weight Loss	25–30%	35–45%	25–30%
Muscle Gain	30–35%	40–50%	20–25%
Ketogenic (if selected)	25–30%	< 5%	65–75%
Diabetes / Pre-diabetes	25–30%	30–40% (low GI only)	25–30%
CKD (Stage 3–4)	0.6–0.8 g/kg body weight	Adjusted to target	Remainder
Endurance / Athletic	20–25%	55–65%	20–25%

Protein targets are always cross-validated against a per-kilogram-body-weight minimum (1.2–2.2 g/kg depending on goal), and the higher of the two values is used.

4. AI Recommendation Engine

The recommendation engine is a four-layer AI pipeline. Each layer processes the outputs of the previous layer and adds reasoning, constraints, or content. The architecture deliberately separates calculation (deterministic), prediction (probabilistic), prescription (optimisation), and memory (adaptive learning).

4.1 Layer 1 — Diagnostic AI: User State Assessment

This layer runs immediately after intake. It analyses the full user profile to produce a structured State Report, which is the input to all downstream layers.

Outputs of Layer 1:

- BMI classification and risk tier.
- Metabolic risk score (0–100) aggregated from BMI, waist circumference, blood glucose, and BP.
- Nutritional gap analysis: comparison of current estimated intake (if provided) vs. computed targets.
- Activity sufficiency score: current activity level vs. WHO minimum recommendations.
- Active constraint flags: list of all health-condition-derived restrictions that must be honoured.
- Readiness-to-train classification: Safe / Conditional / Restricted.

The State Report is logged with a timestamp. Every future week-1 review will diff the new state against the original State Report to quantify progress.

4.2 Layer 2 — Predictive AI: Outcome Forecasting

Before generating a plan, SIKANUA simulates likely outcomes across multiple plan variants. This layer uses a probabilistic model trained on nutritional and exercise science literature to estimate:

- Projected weight change trajectory over 4, 8, and 12 weeks under the proposed caloric target.
- Estimated muscle retention probability under the proposed protein intake.
- Adherence risk score: based on dietary style match, cooking skill, budget, and historical adherence patterns from similar user cohorts.
- Energy availability risk: flags if proposed plan may result in Relative Energy Deficiency in Sport (RED-S) for active users.
- Micronutrient deficiency risk: flags nutrients likely to be low given dietary restrictions (e.g., B12 for vegans, iron for female athletes, calcium for dairy-free users).

Prediction Confidence Levels:

Each prediction is tagged with a confidence level (High / Medium / Low) based on the completeness of the intake profile. Missing Tier-2 or Tier-3 data lowers confidence and widens prediction ranges.

4.3 Layer 3 — Prescriptive AI: Plan Generation

This is the core recommendation layer. It synthesises Layer 1's State Report and Layer 2's forecasts to generate the full 7-day plan. The prescriptive engine operates in three sub-modules:

4.3.1 Diet Plan Module

- Selects meals from a curated, nutritionist-reviewed meal database, filtered by dietary style, allergies, cuisine preferences, and cooking skill.
- Constructs a 7-day meal schedule meeting the computed caloric target and macro distribution within $\pm 5\%$ tolerance.
- Assigns portion sizes dynamically per user (not fixed cookbook portions).
- Generates a consolidated shopping list grouped by store section.
- Flags any days where micronutrient targets are difficult to meet, with specific food suggestions to close the gap.
- Incorporates meal timing guidance based on training schedule (e.g., pre-/post-workout nutrition windows).

4.3.2 Fitness Plan Module

- Selects exercises filtered by equipment access, injury flags, and readiness-to-train classification.
- Constructs a weekly training schedule respecting the user's available days and session duration.
- Applies progressive overload principles for strength plans; periodisation for endurance plans.
- Assigns intensity levels (RPE scale 1–10) based on exercise history and activity level.
- Pairs training days with nutritional support (higher carbs on heavy training days, protein emphasis on rest days).
- Generates video-linked exercise instructions (integration point for exercise media library).

4.3.3 Constraint Enforcement Module

This sub-module applies all active flags as hard constraints before the plan is finalised:

Active Flag	Enforced Constraint
Diabetes	All carbs scored ≤ 55 GI; no added sugar meals; carb distribution across 5–6 small meals
Hypertension	Sodium cap at 1,500 mg/day; no high-sodium condiments; DASH-aligned meals prioritised
CKD (Stage 3–4)	Potassium $< 2,000$ mg/day; phosphorus < 800 mg/day; protein at clinical minimum
Celiac / Gluten-Free	All meals verified gluten-free; oat alternatives specified
Eating Disorder (Recovery)	No calorie counts shown to user; no weight targets displayed; plan framed as nourishment
Cardiac History	Cardio intensity capped at moderate (RPE ≤ 6); no HIIT without clinical clearance
Osteoporosis	Impact exercise limited; resistance training emphasised; calcium and Vitamin D intake flagged
IBS	Low-FODMAP filter applied to meal database; trigger foods excluded by default
PCOS	Anti-inflammatory meals prioritised; refined carbohydrate minimisation; inositol-rich foods suggested

4.4 Layer 4 — Limited Memory AI: Adaptive Learning

After the first 7-day cycle, this layer activates. It ingests the review data (Section 5) and updates the user's internal model to improve subsequent plans. The memory layer maintains:

- Adherence history: per-meal and per-workout completion rate, tracked over all cycles.
- Biometric deltas: weight, body measurements, and any user-reported biometric changes.
- Preference refinements: meals rated highly are upweighted; disliked meals are suppressed.
- Tolerance signals: if a user consistently skips certain meal types, the planner reduces their frequency.
- Response patterns: if the predictive layer's forecasts are consistently off for this user, calibration offsets are applied.
- Workout progression log: weights lifted, distances run, RPE reported — used to auto-progress or regress intensity.

The adaptive layer does not share data across users. All learning is strictly per-user. Anonymised, aggregated trend data may be used to improve the global model in periodic retraining cycles, subject to explicit user consent.

5. The 7-Day Plan Review Protocol

SIKANUA's core operational loop is a rolling 7-day cycle. Every user is placed on a plan for exactly one week before the system triggers a structured review. This cadence balances sufficient time for physiological response with the agility to correct course quickly.

5.1 Review Trigger Mechanism

- The review is automatically triggered at 23:59 on Day 7 of each plan cycle, regardless of adherence level.
- Users receive a push notification at Day 5 previewing that a review is upcoming and prompting them to log any outstanding meals or workouts.
- Emergency early review is triggered if: the user logs a health symptom (dizziness, chest pain, extreme fatigue); weight drops more than 2 kg in one week; the user manually requests a review.

5.2 Review Data Collection

At the end of each 7-day cycle, the system collects the following data points either automatically (from app logs) or through the user's end-of-week check-in:

5.2.1 Adherence Metrics (Auto-collected)

- Meals logged as completed vs. total planned meals.
- Workouts logged as completed vs. total planned workouts.
- Water intake average (if tracked).
- Sleep duration average (if integrated with wearable).

5.2.2 Biometric Updates (User-reported or Device-synced)

- Current weight.
- Body measurements if tracked (waist, hip).
- Subjective energy level: 1–10 scale.
- Subjective hunger level: 1–10 scale (averaged across week).
- Any new symptoms or health changes.

5.2.3 Satisfaction & Feedback

- Overall plan satisfaction: 1–5 star rating.
- Diet plan satisfaction: 1–5 star rating with optional notes.
- Fitness plan satisfaction: 1–5 star rating with optional notes.
- Specific meals to keep, modify, or remove.
- Specific exercises to keep, modify, or remove.
- Open-text field: anything else you want us to know.

5.3 Review Decision Matrix

The Layer 4 adaptive engine processes the review data and produces one of five continuation decisions:

Decision	Trigger Conditions	Action Taken
Continue Unchanged	≥80% adherence; biometrics on track; satisfaction ≥4	Regenerate same plan structure for next 7 days with minor variety swaps
Minor Adjustment	60–79% adherence OR satisfaction 3–4 OR slight biometric deviation	Swap disliked meals; adjust one training day; recalibrate portions
Moderate Revision	<60% adherence OR satisfaction ≤2 OR biometrics not trending correctly	Rebuild diet plan; revise workout intensity; re-run predictive layer
Full Rebuild	Combination of low adherence + poor satisfaction + adverse biometrics	Full new plan generation as if new user, incorporating all memory data
Clinical Escalation	Any red-flag symptom logged; BMI drop > 1.5 in one week; health flags triggered	Plan paused; user routed to clinician review queue; no new plan until cleared

5.4 Review Communication to User

After the decision is made, the user receives a Review Summary notification containing:

- A plain-language progress summary (no clinical jargon).
- One highlight ('What went well this week').
- One focus area ('What to build on next week').
- The next 7-day plan, ready to start immediately.
- An optional 'Chat with Advisor' prompt if they want to discuss the plan.

Communication Principle: The review summary is framed positively and constructively at all times. The system never uses language that implies failure, guilt, or punishment for low adherence. Low adherence is treated as a signal to adjust the plan, not a character assessment of the user.

6. Safety Architecture & Guardrails

SIKANUA is a health system. Safety guardrails are not optional features — they are structural constraints built into every layer of the engine. This section documents all safety mechanisms.

6.1 Caloric & Nutritional Safety Floors

- Minimum daily calories: 1,200 kcal (female users), 1,500 kcal (male users). Hard-coded floor; cannot be overridden by any input combination.
- Minimum protein: 0.8 g/kg/day for general users; higher minimums enforced for active users and those with muscle-preservation goals.
- Maximum deficit: 1,000 kcal/day below TDEE. Plans requiring a larger deficit to meet a user's timeframe goal will extend the timeframe instead.
- Micronutrient floor: if any day's plan falls below 50% of the RDA for any tracked micronutrient without a supplement compensating, the meal is replaced.

6.2 Exercise Safety Limits

- RPE cap: no first-week plan exceeds RPE 7. The system ramps intensity over weeks 2–4.
- Cardiac safety: any user with a cardiac history flag is capped at moderate-intensity cardio and must clear HIIT with a clinician first.
- Progressive overload limit: strength plan increases volume by no more than 10% per week (standard progressive overload safety principle).
- Rest day enforcement: every 7-day plan must include at least 1 full rest day and 1 active recovery day.

6.3 Clinical Escalation Triggers

The following conditions immediately pause the active plan and route the user to a supervised queue:

- User logs: chest pain, shortness of breath, severe dizziness, fainting, or severe nausea during exercise.
- Logged weight change exceeds -2 kg in a 7-day period.

- Blood glucose log (if connected) records a reading < 3.9 mmol/L (hypoglycaemia threshold).
- Eating disorder recovery flag becomes active mid-plan.
- User-initiated distress signal via in-app 'I need help' button.

6.4 Data Privacy Guardrails

- All health condition data is stored encrypted at rest and in transit.
- Biometric data is never shared with third parties without explicit, granular user consent.
- The adaptive learning model is trained per-user; no cross-user data inference.
- Users may request deletion of all data at any time; deletion cascades to the adaptive memory store.
- GDPR, HIPAA, and applicable local health data regulations compliance is mandatory before production deployment.

6.5 Equity & Bias Monitoring

SIKANUA is designed to serve a diverse global population. The following bias monitoring protocols are in place:

- Meal databases are curated to include regional cuisine representation from Africa, Asia, Latin America, the Middle East, and the Caribbean — not defaulted to Western diets.
- BMI thresholds for Asian populations are adjusted per WHO Asia-Pacific guidelines (overweight threshold: 23.0; obese threshold: 27.5).
- All recommendation outputs are reviewed quarterly across demographic cohorts to detect and correct disparate outcomes.
- The system never penalises or downgrades the quality of recommendations based on budget category.

7. End-to-End System Workflow

The following outlines the full user journey through SIKANUA from initial signup through the ongoing weekly plan cycle:

7.1 Onboarding Flow (One-Time)

1. User registers and completes Tier-1 intake form (biometrics, goal, dietary style, activity level).
2. System computes BMR, TDEE, and caloric target (Baseline Calculation Engine, Section 3).
3. Layer 1 Diagnostic AI produces the State Report.
4. Layer 2 Predictive AI forecasts outcomes and flags adherence risks.
5. Constraint Enforcement Module applies all active health flags.
6. Layer 3 Prescriptive AI generates the first 7-day Diet + Fitness Plan.
7. Clinical safety check: if any escalation trigger is active, plan is held and user is queued.
8. Plan is delivered to the user via the app dashboard.

7.2 Weekly Operational Cycle

- 9. User follows the 7-day plan; meals and workouts are logged in-app.
- 10. Day 5: Preview notification prompts logging completion.
- 11. Day 7 (23:59): Review is triggered automatically.
- 12. End-of-week check-in form is presented; user submits biometric updates and feedback.
- 13. Layer 4 Memory AI ingests all review data and computes adherence, biometric delta, and satisfaction scores.
- 14. Review Decision Matrix produces one of five decisions (Section 5.3).
- 15. Layer 3 regenerates or revises the next 7-day plan accordingly.
- 16. Review Summary is delivered to the user; new plan begins immediately.
- 17. Repeat from Step 1.

7.3 Long-Term Progression Logic

SIKANUA adjusts its approach based on cumulative plan cycles:

Cycle	Behaviour
Week 1–2	Conservative: lower intensity, familiar foods, high variety to identify preferences
Week 3–4	Progression: intensity ramps, macro targets tightened based on adherence response
Week 5–8	Optimisation: full adaptive personalisation active; meal and workout preferences well-calibrated
Week 9–12	Performance: system shifts from learning to optimising — fine-tuning rather than rebuilding
12+ Weeks	Maintenance or goal transition: if primary goal is met, system prompts goal re-assessment

8. Integration Points & External Data Sources

8.1 Wearable & Health Device Integrations

SIKANUA supports the following data integrations, all of which are optional and consent-gated:

- Apple Health / Google Fit: steps, active calories, sleep, heart rate.
- Garmin / Fitbit / Whoop: workout tracking, recovery scores, HRV.
- Continuous Glucose Monitors (CGM): real-time blood glucose data for diabetic users.
- Smart Scales: weight and body composition sync.

8.2 Meal Database & Food API

- Primary food database: USDA FoodData Central (global fallback).
- Regional supplement databases: West Africa Food Composition Table, ASEAN Food Composition Database.
- Barcode scanning integration for packaged food logging.
- Restaurant menu API integration (where available) for eating-out logging.

8.3 Clinician Portal

Users escalated to the clinical queue are managed through a separate Clinician Portal. The portal exposes:

- Full user State Report and plan history.
- Escalation reason and flag detail.
- Clinician override: ability to modify constraints, approve plans, or mark a case as cleared.
- Secure messaging channel between clinician and user.
- Audit log of all clinician actions.

9. Glossary

Term	Definition
BMR	Basal Metabolic Rate — energy expended at complete rest
TDEE	Total Daily Energy Expenditure — BMR adjusted for activity level
BMI	Body Mass Index — weight in kg divided by height in metres squared
PAL	Physical Activity Level — multiplier applied to BMR to estimate TDEE
RPE	Rate of Perceived Exertion — subjective 1–10 exercise intensity scale
DASH	Dietary Approaches to Stop Hypertension — evidence-based dietary pattern
GI	Glycaemic Index — ranking of carbohydrate-containing foods by blood glucose response
RED-S	Relative Energy Deficiency in Sport — condition from insufficient energy intake in active individuals
FODMAP	Fermentable Oligosaccharides, Disaccharides, Monosaccharides And Polyols — carb group linked to IBS
CKD	Chronic Kidney Disease — staged kidney function impairment affecting dietary requirements
PCOS	Polycystic Ovary Syndrome — hormonal condition affecting metabolic and dietary needs
RDA	Recommended Dietary Allowance — established minimum intake for specific nutrients
State Report	SIKANUA's internal diagnostic summary of a user's current health and biometric profile
Plan Cycle	A single 7-day diet and fitness plan period, ending in a review

Term	Definition
Constraint Flag	A health-condition-derived rule that limits or modifies plan options